Ex. No :08 **App Architecture – UI Layer**

Aim

To understand the UI layer in Android architecture by handling user interaction and updating UI using Activity and ViewModel.

Procedure

1. Open Android Studio.
2. Create a new project (Empty Activity, Kotlin).
3. Design UI in activity_main.xml using:
    - TextView (to display message)
    - Button (to trigger action)
4. Create a **ViewModel class** to manage data.
5. Open MainActivity.kt.
6. Connect UI components using findViewById.
7. Use ViewModel to fetch data.
8. Handle button click and update UI.
9. Run the app and observe output.

Code

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:orientation="vertical"
   android:padding="20dp"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <TextView
      android:id="@+id/tvTitle"
      android:text="UI Layer Demo"
      android:textSize="22sp"
      android:gravity="center"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"/>

   <TextView
      android:id="@+id/tvMessage"
      android:text="Message will appear here"
      android:textSize="18sp"
      android:paddingTop="20dp"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"/>
```

```
    <Button
        android:id="@+id/btnLoad"
        android:text="Load Data"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

MainViewModel.kt

```
package com.example.uilayerdemo

import androidx.lifecycle.ViewModel

class MainViewModel : ViewModel() {

    fun getMessage(): String {
        return "Hello from ViewModel"
    }
}
```

MainActivity.kt

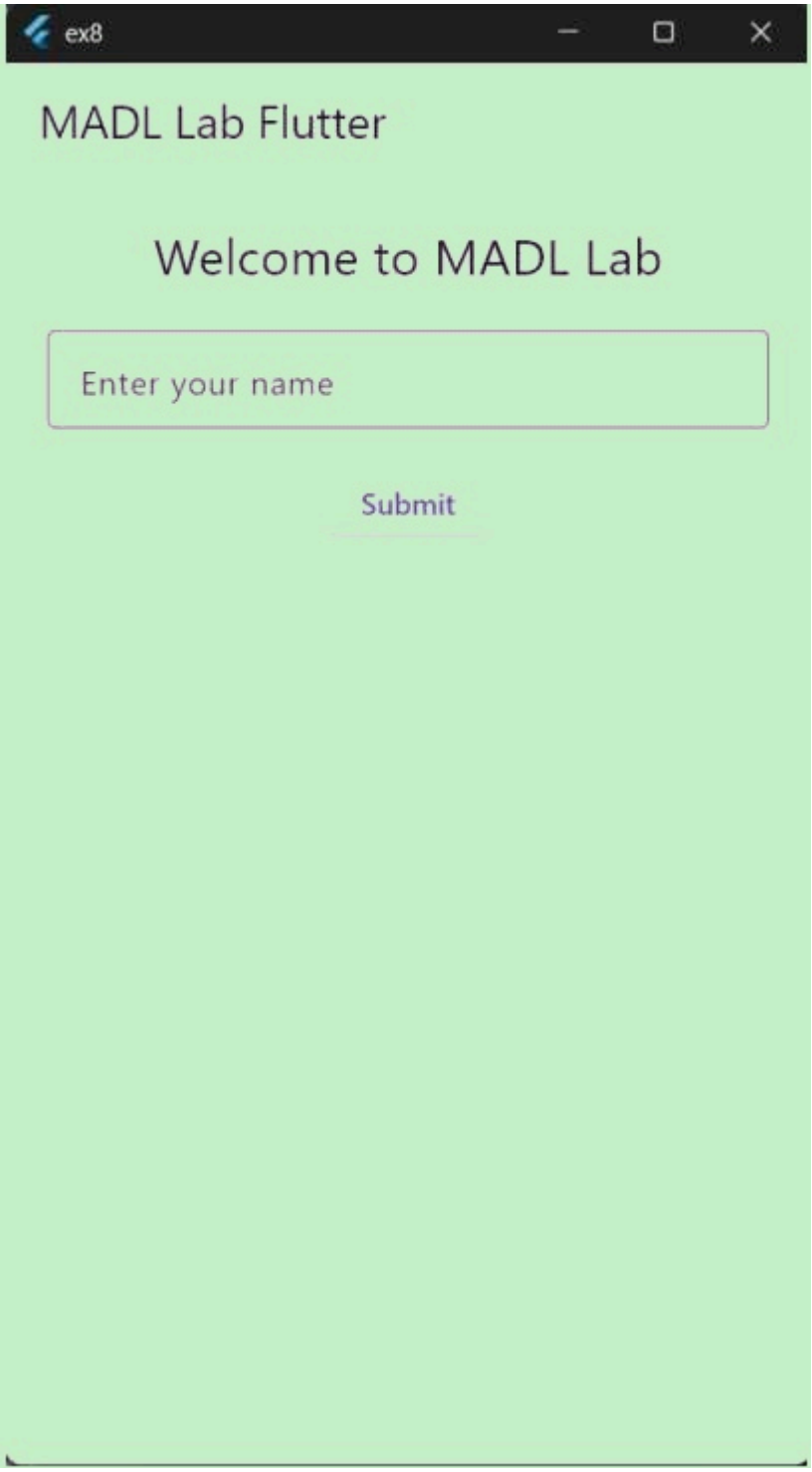
```
package com.example.uilayerdemo

import android.os.Bundle
import android.widget.Button
import android.widget.TextView
import androidx.activity.viewModels
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    private val viewModel: MainViewModel by viewModels()

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val tvMessage = findViewById<TextView>(R.id.tvMessage)
        val btnLoad = findViewById<Button>(R.id.btnLoad)

        btnLoad.setOnClickListener {
            tvMessage.text = viewModel.getMessage()
        }
    }
}
```

Result

The application successfully:

- Handles user interaction through UI layer
- Uses ViewModel to provide data

- Displays dynamic message on button click



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